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CXR-ViT: A Transformer-Based Framework for Enhanced Chest X-Ray Classification

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***Abstract*—Increasing need for accurate medical image analysis, chest X-ray classification has become a powerful tool for diagnosing pulmonary diseases. However, the inherent complexity of thoracic abnormalities and the limited annotated medical data pose significant challenges to accurate classification. In this study, we utilized fine-tuned deep learning-based CNN and large vision transformer frameworks to classify lung diseases from CXR images. Moreover, to address the issues of model interpretability and decision-making, we use Gradient-weighted Class Activation Map (Grad-CAM) to visualize the results and improve the model predictions' interpretability and accuracy by identifying relevant regions in the lung areas. The results of our experiments on the publicly available COVID-QU-Ex dataset show that the proposed CXR-ViT outperforms the conventional CNN approaches with an accuracy of 99.01 %. This work highlights the use of transformers in medical image analysis, emphasizing better lung disease classification and interpretability.**

Keywords—CXR-ViT, Chest X-ray, Covid-19, Transfer Learning, CNN, Vision Transformer, Grad-CAM, COVID-QU Ex dataset.

I. INTRODUCTION

Deep learning (DL) has revolutionized medical image analysis due to its exponential rate of development and better results in tasks like diagnosis, segmentation, and classification [1, 2]. However, a fundamental problem is that there are fewer annotated medical imaging datasets, which is an essential factor in ensuring that the DL models generalize well [3, 4]. Transfer learning has become a significant method for solving this issue in medical image analysis [5- 7]. A CNN trained originally on an extensive image database (such as the ImageNet) is then fine-tuned on a comparatively small medical image dataset with the motive that the features extracted from natural images facilitate the CNNs to address DL tasks with reasonable accuracy [8-10].

In this direction, Ahsan et al.[11] used transfer learning with the VGG16 model to diagnose TB from Chest X-ray (CXR) images. After fine-tuning the model with a dataset of 1,324 CXR images, the study attains an accuracy of 80% in differentiating TB images from healthy images. Similarly, shelke et al. [12] employed three pre-trained CNN models for different classifications using a different approach. Timemy et al. [13] used 14 pre-trained CNN models for feature extraction, and 5-fold cross-validation was done on the database of 2186 CXR images, yielding a detection accuracy of 91.6%. Another study by Shastri et al. [14] utilized a pre-trained Inception-V3 model and transfer learning on 3532 improved CXR images, with each image reduced to 299 × 299 × 3 bits' resolution to gain a 93% accuracy. However, this study did not involve categorically identifying TB in the CXR images. In another work, Rahimzadeh et al. [15] opted for multiple feature extraction through parallel processes to support classification and achieved an accuracy of 91.4%.

One of the main drawbacks of these studies is that they use models built under the CNN framework, which has a limited receptive field and was initially designed for local feature extraction. This characteristic limits CNNs in capturing long-range dependencies and global context information in medical images such as chest X-rays, where specific diagnostic patterns may cover more significant regions. To overcome these limitations, large-scale transformer models trained on natural image data, known as vision transformers (ViTs), have recently been proposed and demonstrated superior performance in a range of image processing benchmarks than conventional CNN structures. Due to the relatively large receptive field of the transformer, compared to that of CNNs, the transformer's proficiency in recognizing the global context in images has been another reason that makes it an appealing model for medical image analysis, especially when data is scarce, such as Sitaula et al. [16], used VGG16 and attention mechanism to classify CXR images using three CXR datasets and achieved an accuracy of 87.49%.

This work analyzes the suitability of fine-tuned CXR-ViT for medical image classification and, more specifically, CXR images. In this work, we utilize the COVID-QU-Ex dataset and evaluate the performance of the proposed fine-tuned CXR-ViT model alongside fine-tuned CNN models: VGGNet, EfficientNet, DenseNet, and Res-Net. Finally, we intend to compare the results of each model to learn if fine-tuned vision transformers can excel in transfer learning compared to CNNs in medical image processing. Our experiments showed that the proposed fine-tuned CXR-ViT model performs better than CNN-based architectures by showing enhanced transfer learning capability. Thus, the transformers possess a high potential for developing the area of medical image classification, given that the availability of large-scale labeled datasets often remains a concern.

The major contribution of our study is summarized as:

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— Proposed a fine-tuned chest X-ray vision transformer (CXR-ViT) framework for classifying lung diseases, addressing the limitations of CNNs in capturing long-range dependencies and global context.

— Integrated Grad-CAM to visualize infected lung regions, improving model interpretability and supporting informed clinical decisions.

— Comprehensively compared the proposed CXR-ViT model with fine-tuned benchmark CNN architectures to highlight the advantages of proposed approach.

The remainder of this section is organized as follows. Section 2 discusses materials and methods, section 3 presents results and discussion, and section 4 discusses the conclusion.

II. MATERIALS AND METHODS

A. Dataset Description

The COVID-QU-Ex dataset (as shown in TABLE I) is sourced from Qatar University and comprises 33,920 CXR images [17, 18]. This dataset is categorized into three distinct classes: COVID-19, non-COVID infections (viral and bacterial pneumonia), and normal. Due to the extensive characteristics of this dataset, it can provide substantial support to diagnose and classify lung diseases for enhancing accuracy.

TABLE I. CLASS COMPOSITION OF COVID-QU-EX DATASET IMAGES [17, 18].

Disease classes	Training set	Testing set
Covid-19	8369	3587
Non-covid	7884	3379
Normal	7490	3211

B. Proposed fine-tuned CXR-ViT Architecture

In this study, we develop a fine-tuned Vision Transformer (CXR-ViT) model (shown in Fig.1) for classifying lung diseases, including COVID-19, non-COVID, and normal cases. The model utilizes a self-attention mechanism to capture dependencies between image patches. This increases its ability to detect subtle patterns in medical images. The fine-tuning is achieved by freezing the initial layers of the model. These layers retain the generic visual patterns learned from the ImageNet dataset, which are transferable across different datasets. Freezing these layers ensures that their pre-trained knowledge is preserved and saves training time. Subsequently, we replace the model's original classification head with a new custom head optimized specifically for lung disease classification. This new head includes dense layers for representation learning, dropout layers to mitigate overfitting, and a SoftMax layer to output a probability distribution over the disease classes. By incorporating this custom head, the model is capable of estimating class probabilities for the given input images. In the next phase, we selectively unfreeze some of the deeper layers of the model. These layers, closer to the output, are responsible for learning task-specific representations. Unfreezing them allows the model to adapt better to the features of the lung disease dataset. To prevent drastic changes in the pre-trained feature space, we use a small learning rate during this fine-tuning step. This approach enables the model to acquire new, domain-specific features relevant to lung disease classification while maintaining the knowledge gained from pretraining.

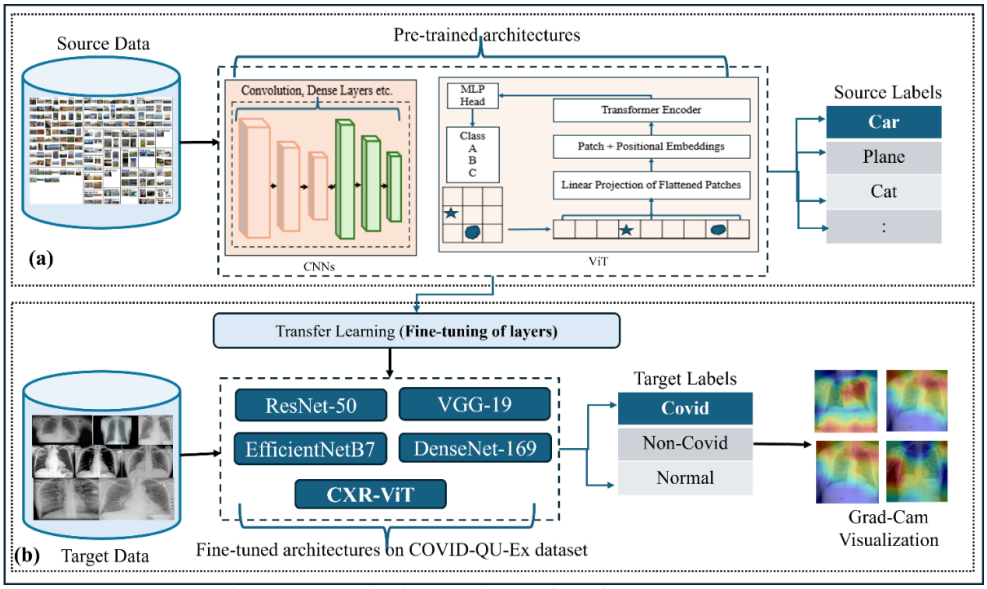


Fig. 1: (a) conventional process (b) workflow of the proposed study.

For optimization, we use the Adam optimizer along with the categorical cross-entropy loss function to measure the difference between predicted and actual labels. Early stopping is also applied to prevent overfitting and ensure efficient training. To further enhance the model's performance, we use Batch Normalization layers. These layers normalize activations at each layer, improving training stability and accelerating convergence. Finally, to improve the interpretability of the

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model, we apply the Grad-CAM technique. This allows us to identify the regions of the input images that the model focuses on during classification. This step helps ensure that the model's decisions are more transparent and clinically relevant.

Besides the proposed model, we also evaluated the performance of benchmark CNN models including ResNet-50 [19], DenseNet-169 [20], VGG-19 [21], EfficientNetB7 [22], and Vision Transformer (ViT) [23] to assess the effectiveness in classifying lung diseases. These models are considered for different reasons such as ResNet-50 provides high-quality features from residual connections, DenseNet-169 has dense layer connectivity, VGG-19 is easy to understand, EfficientNetB7 maintains a balance between scaling and other factors, ViT has a self-attention mechanism.

C. Performance Metrics

The performance of the proposed model and other benchmark models are evaluated using the four evaluation metrics as described in Eq.1 to Eq.4 [24,25,26,27]. The term terms TP, TN, FP, FN denotes the true positive, true negative, false positive, and false negative, respectively. Where, P=TP+FN and N=TN+FP.

$$\text{Accuracy} = \frac{TP+TN}{P+N} \times 100 \quad (1)$$

$$\text{Precision} = \frac{TP}{TP+FP} \times 100 \quad (2)$$

$$\text{Recall} = \frac{TP}{P} \times 100 \quad (3)$$

$$\text{F1-Score} = \frac{2 \times precision \times recall}{precision + recall} \times 100 \quad (4)$$

III. RESULTS AND DISCUSSION

The result of the classification of CXR images using the VGG-19, ResNet-50, DenseNet-169, EfficientNetB7, and ViT models show that the model's variance in the accuracy, precision, recall, and F1-score measurements is due to the architecture of each model (as shown in TABLE II). The proposed CXR-ViT achieved the highest accuracy of 99.01% compared to the other state-of-the-art CNN models. This result can be attributed to its attention-based structure, which allows the model to learn the inherent image dependencies. The EfficientNetB7 reached the second-highest top accuracy of 97.93% with its compound scaling to enhance the depth, width, and resolution for better feature extraction. In DenseNet-169 and ResNet-50, we additionally observed high accuracies of 97.31% and 96.23%, primarily due to the dense connectivity in DenseNet and residual connections in ResNet but slightly lowered by EfficientNetB7's scaling. Nevertheless, the lowest accuracy was achieved by the VGG-19 model at 95.45% due to the relatively simple connectivity and less complex scaling for capturing intricate features in medical images. Precision, which indicates how accurately the model identifies the positive cases among the predicted positive ones, was the highest for proposed CXR-ViT, with 98.52%. The detection accuracy of EfficientNetB7 came next with a score of 96.86%, thus proving that balanced scaling also reduces false cases efficiently. DenseNet-169 and ResNet-50 had high precision at 95.95% and 94.18% respectively. DenseNet-169 model maintains high precision due to its densely connected layers outperforming ResNet in precision. VGG-19 achieved the least precision of 92.84% and showed that the sequential arrangement was still incapable of producing high class-specific features. Proposed CXR-ViT also had a significantly higher recall rate at 98.68%, proving that it can capture true positives and minimize false negatives, which is critical in many medical applications where a significant number of actual positives should not be missed. EfficientNetB7 achieved a recall of 97.29%, showing that the model is highly sensitive to the true positives. DenseNet-169, with a recall of 96.45%, also achieved a good score because of its feature reuse capability. ResNet-50 and VGG-19 achieved 95.20% and 94.36% recall, respectively, which is less compared to other models, indicating that these models can misguide by labeling some of the positive cases as negative due to less complex connectivity in ResNet or due to absence of residual connections in VGG-19. The F1-score, which combines precision and recall, was the highest in the proposed CXR-ViT at 98.59%, the second highest in Efficient-NetB7 at 97.07%, while DenseNet-169 had the least at 96.19%. The high F1 scores of the models mirror the relative balance between precision and recall: ViT benefits from its capacity to capture global dependencies, EfficientNetB7 due to its scaling strategy, and DenseNet-169 due to connected features' propagation. ResNet-50 obtains an F1-score (94.68%) lower than DenseNet models because ResNet bypasses features and propagates them differently, so it does not have the same level of feature propagation as DenseNet models. VGG-19 got the lowest F1 score (93.59%) due to its sequential structure, which does not allow deep feature extraction. The Vision Transformer yielded the best overall performance concerning all the considered metrics, indicating the robustness of self-attention for distinguishing between medical images. The results show that the models that use advanced connectivity, efficient scalability, or attention-based aspects have higher accuracy, precision, recall, and F1 scores, making them more suitable for diagnostic tasks such as CXR classification.

TABLE II. PERFORMANCE OF VARIOUS MODELS FOR CXR CLASSIFICATION.

Models used	Accuracy	Precision	Recall	F1-Score
VGG-19	95.45	92.84	94.36	93.59
ResNet-50	96.23	94.18	95.20	94.68
DenseNet-169	97.31	95.95	96.45	96.19
EfficientNetB7	97.93	96.86	97.29	97.07
CXR-ViT	**99.01**	**98.52**	**98.68**	**98.59**

The graph shown in Fig 2 compares the training and validation loss of the proposed CXR-ViT model over 100 Epochs. It explains how the model is learning and how well it can generalize. Both loss values are high for the first epochs, implying that the model is still in the learning phase in epochs 0–20. At this stage, the model parameters are adjusted, and the resulting training and validation loss curves start falling. When training is over, a clear trend of the decrease in both training and validation losses can be seen. However, the training loss is

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smoother, and a slight fluctuation in validation loss is justified since a model is tested on a validation dataset that it has not seen before. After approximately 20 epochs, the validation loss remains generally above the training loss, indicating that the model is slightly overfitting on the training data. Nonetheless, loss values become stable and are close to each other at the end of the learning process, especially in the last 80-100 epochs, indicating good generalization ability. The loss curves imply an excellent learning progression throughout the epochs, indicating that the proposed CXR-ViT can perform well for CXR classification.

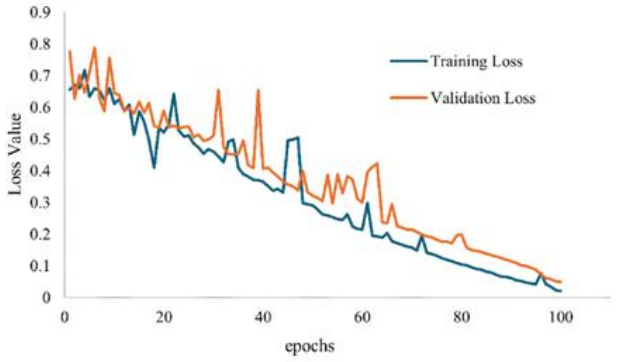


Fig. 2: Training and validation loss of proposed CXR-ViT over 100 epochs.

The Grad-CAM heatmap visualization of deeply extracted features responsible for disease detection in CXR images is illustrated in Fig. 3. The evaluation of the proposed CXR-ViT model shows more accurate and localized attention on the lung regions that are relevant to the COVID-19 category and the other classes than the different models. We observed that VGG-19 and ResNet-50 generate activations of the images at a general level with little focus on specific regions of the CXR. The proposed CXR-ViT, being highly focused and regional, locates the areas of interest more accurately in distinguishing lung features.

TABLE III. PERFORMANCE OF VARIOUS MODELS FOR CXR CLASSIFICATION.

Study	Model used	Accuracy
Kamal et al. [28]	XceptionNet	98.86%
Kumar et al. [30]	CNN Ensemble	95.00%
Pacal et al. [31]	MaxViT	97.14%
Baker et al. [29]	XceptionNet	98.13%
Proposed Method	**CXR-ViT**	**99.01%**

TABLE III gives a comparative analysis of the recent DL models for CXR classification in terms of accuracy. Kamal et al. [28] and Baker et al. [29] used the XceptionNet model and achieved approximately—98%, reflecting the effectiveness in medical image classification. Kumar et al. [30] used a CNN ensemble with an accuracy of 95%. Pacal et al. [31] employed the MaxViT model and achieved an accuracy of 97.14%. The CXR-ViT model proposed in this study attained a maximum accuracy of 99.01%, showing that Vision Transformers can be significantly accurate in analyzing lung diseases.

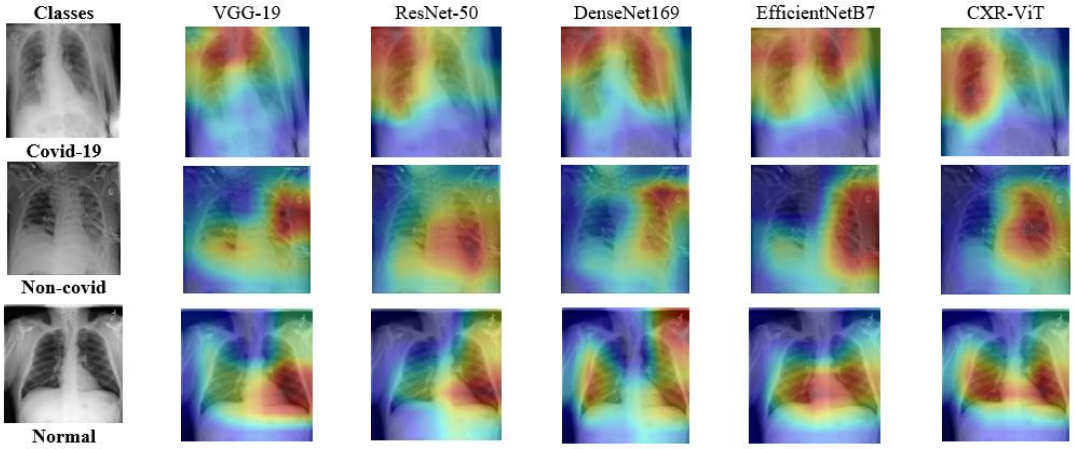


Fig. 3: Grad-Cam visualization instances of COVID-19, non-covid, and normal.

IV. CONCLUSION

The empirical results presented in this study substantiate the efficacy of the proposed CXR-ViT model as a robust approach for advancing lung disease classification. The COVID-QU-Ex dataset used in this research shows that transfer learning helps the CXR-ViT model outperform traditional CNN-based frameworks with an accuracy of 99.01%. The application of Grad-CAM resulted in improved interpretability of model decision-making and demonstrated high confidence in highlighting the critical lung areas related to different pulmonary diseases. These results indicate that trans-former-based models can be a valuable and versatile technique in medical imaging, making it easy to classify CXR images. This work highlights how the transformer can be better implemented in the medical field, particularly for challenging classification tasks with few labeled data. Future work might include increasing the sample size, optimizing the model architectures, and analyzing more advanced methods of interpreting the results to improve accuracy and credibility.

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REFERENCES

[1] Medeiros, E.P., Machado, M.R., de Freitas, E.D.G., da Silva, D.S., de Souza, R.W.R.: Applications of machine learning algorithms to support COVID-19 diagnosis using X-rays data information. Expert Syst Appl. 238, 122029 (2024). https://doi.org/10.1016/j.eswa.2023.122029.

[2] Kumar, S., Bhagat, V., Sahu, P., Chaube, M.K., Behera, A.K., Guizani, M., Gravina, R., Di Dio, M., Fortino, G., Curry, E., Alsamhi, S.H.: A novel mul-timodal framework for early diagnosis and classification of COPD based on CT scan images and multivariate pulmonary respiratory diseases. Comput Methods Programs Biomed. 243, 107911 (2024). https://doi.org/10.1016/j.cmpb.2023.107911.

[3] Alshehri, E., Kalkatawi, M., Abukhodair, F., Khashoggi, K., Alotaibi, R.: COVID-19 Diagnosis from Medical Images Using Transfer Learning. Saudi Journal of Health Systems Research. 2, 54–61 (2022). https://doi.org/10.1159/000521658.

[4] Kansal, K., Chandra, T. B., & Singh, A. (2024). Advancing differential diagnosis: a comprehensive review of deep learning approaches for differentiating tuberculosis, pneumonia, and COVID-19. Multimedia Tools and Applications. https://doi.org/10.1007/s11042-024-19350-1

[5] Chandra, T. B., Singh, B. K., & Jain, D. (2022). Disease Localization and Severity Assessment in Chest X-Ray Images using Multi-Stage Superpixels Classification. Computer Methods and Programs in Biomedicine, 222, 106947. https://doi.org/10.1016/j.cmpb.2022.106947

[6] Chandra, T. B., Verma, K., Singh, B. K., Jain, D., & Netam, S. S. (2020). Automatic detection of tuberculosis related abnormalities in Chest X-ray images using hierarchical feature extraction scheme. Expert Systems with Applications, 158, 113514. https://doi.org/10.1016/j.eswa.2020.113514

[7] Khalid, S., Rashwan, H. A., Abdulwahab, S., Abdel-Nasser, M., Quiroga, F. M., & Puig, D. (2024). FGR-Net: Interpretable fundus image gradeability classification based on deep reconstruction learning. Expert Systems with Applications, 238, 121644. https://doi.org/10.1016/j.eswa.2023.121644

[8] Medeiros, E. P., Machado, M. R., de Freitas, E. D. G., da Silva, D. S., & de Souza, R. W. R. (2024). Applications of machine learning algorithms to support COVID-19 diagnosis using X-rays data information. Expert Systems with Applications, 238, 122029. https://doi.org/10.1016/j.eswa.2023.122029

[9] Ding, K., Chen, S., Zeng, Y., Liu, Y., Xu, B., & Wang, Y. (2024). SDTU-Net: Stepwise-Drop and Transformer-Based U-Net for Subject-Sensitive Hashing of HRRS Images. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 17, 3836–3849. https://doi.org/10.1109/JSTARS.2024.3356660

[10] Kumar, S., Bhagat, V., Sahu, P., Chaube, M. K., Behera, A. K., Guizani, M., Gravina, R., Di Dio, M., Fortino, G., Curry, E., & Alsamhi, S. H. (2024). A novel multimodal framework for early diagnosis and classification of COPD based on CT scan images and multivariate pulmonary respiratory diseases. Computer Methods and Programs in Biomedicine, 243, 107911. https://doi.org/10.1016/j.cmpb.2023.107911

[11] Ahsan, M., Gomes, R., Denton, A.: Application of a Convolutional Neural Network using transfer learning for tuberculosis detection. In: 2019 IEEE In-ternational Conference on Electro Information Technology (EIT). pp. 427–433. IEEE (2019). https://doi.org/10.1109/EIT.2019.8833768.

[12] Shelke, A., Inamdar, M., Shah, V., Tiwari, A., Hussain, A., Chafekar, T., Me-hendale, N.: Chest X-ray Classification Using Deep Learning for Automated COVID-19 Screening. SN Comput Sci. 2, 300 (2021). https://doi.org/10.1007/s42979-021-00695-5.

[13] Al-Timemy, A.H., Khushaba, R.N., Mosa, Z.M., Escudero, J.: An Efficient Mixture of Deep and Machine Learning Models for COVID-19 and Tubercu-losis Detection Using X-Ray Images in Resource Limited Settings. Presented at the (2021). https://doi.org/10.1007/978-3-030-69744-0_6.

[14] Shastri, S., Kansal, I., Kumar, S., Singh, K., Popli, R., Mansotra, V.: CheXI-mageNet: a novel architecture for accurate classification of Covid-19 with chest x-ray digital images using deep convolutional neural networks. Health Technol (Berl). 12, 193–204 (2022). https://doi.org/10.1007/s12553-021-00630-x.

[15] Rahimzadeh, M., Attar, A.: A modified deep convolutional neural network for detecting COVID-19 and pneumonia from chest X-ray images based on the concatenation of Xception and ResNet50V2. Inform Med Unlocked. 19, 100360 (2020). https://doi.org/10.1016/j.imu.2020.100360.

[16] Sitaula, C., Hossain, M.B.: Attention-based VGG-16 model for COVID-19 chest X-ray image classification. Applied Intelligence. 51, 2850–2863 (2021). https://doi.org/10.1007/s10489-020-02055-x.

[17] Tahir, A.M., Chowdhury, M.E.H., Khandakar, A., Rahman, T., Qiblawey, Y., Khurshid, U., Kiranyaz, S., Ibtehaz, N., Rahman, M.S., Al-Maadeed, S., Mahmud, S., Ezeddin, M., Hameed, K., Hamid, T.: COVID-19 infection local-ization and severity grading from chest X-ray images. Comput Biol Med. 139, 105002 (2021). https://doi.org/10.1016/j.compbiomed.2021.105002.

[18] Rahman, T., Khandakar, A., Qiblawey, Y., Tahir, A., Kiranyaz, S., Abul Ka-shem, S. Bin, Islam, M.T., Al Maadeed, S., Zughaier, S.M., Khan, M.S., Chowdhury, M.E.H.: Exploring the effect of image enhancement techniques on COVID-19 detection using chest X-ray images. Comput Biol Med. 132, 104319 (2021). https://doi.org/10.1016/j.compbiomed.2021.104319.

[19] Chen, Y., Lin, Y., Xu, X., Ding, J., Li, C., Zeng, Y., Liu, W., Xie, W., Huang, J.: Classification of lungs infected COVID-19 images based on inception-ResNet. Comput Methods Programs Biomed. 225, 107053 (2022). https://doi.org/10.1016/j.cmpb.2022.107053.

[20] Huang, G., Liu, Z., Van Der Maaten, L., Weinberger, K.Q.: Densely Connect-ed Convolutional Networks. In: 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 2261–2269. IEEE (2017). https://doi.org/10.1109/CVPR.2017.243.

[21] Shibu George, G., Raj Mishra, P., Sinha, P., Ranjan Prusty, M.: COVID-19 detection on chest X-ray images using Homomorphic Transformation and VGG inspired deep convolutional neural network. Biocybern Biomed Eng. 43, 1–16 (2023). https://doi.org/10.1016/j.bbe.2022.11.003.

[22] Howard, A.G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., Andreetto, M., Adam, H.: MobileNets: Efficient Convolutional Neural Net-works for Mobile Vision Applications. (2017).

[23] Hu, Z., Wang, Z., Jin, Y., Hou, W.: VGG-TSwinformer: Transformer-based deep learning model for early Alzheimer's disease prediction. Comput Methods Programs Biomed. 229, 107291 (2023). https://doi.org/10.1016/j.cmpb.2022.107291.

[24] Lalini Chandra, Mithilesh Atulkar, and Priyanka Tripathi. 2024. Fusion of Local and Global Texture Descriptors for Improved Tuberculosis Detection in Chest X-ray Images. In 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT), 1–7. https://doi.org/10.1109/ICCCNT61001.2024.10724538

[25] Kajal Kansal, Tej Bahadur Chandra, and Akansha Singh. 2024. ResNet-50 vs. EfficientNet-B0: Multi-Centric Classification of Various Lung Abnormalities Using Deep Learning. Procedia Computer Science 235: 70–80. https://doi.org/10.1016/j.procs.2024.04.007

[26] Kajal Kansal, Tej Bahadur Chandra, Akansha Singh, and Krishna Kant Singh. 2024. E-CNN: ensembled CNN learning approach for pneumonia detection in chest X-ray images. In IET Conference Proceedings, 80–86. https://doi.org/10.1049/icp.2024.2532

[27] Lipismita Panigrahi, Tej Bahadur Chandra, Atul Kumar Srivastava, Neeraj Varshney, Kamred Udham Singh, and Shambhu Mahato. 2024. mBCCf: Multilevel Breast Cancer Classification Framework Using Radiomic Features. International Journal of Intelligent Systems 2024: 1–24. https://doi.org/10.1155/2024/6631016

[28] Kamal, S., Sarfraz, M., Fatma, S., Al Fatma, I., Parween, S., Al Maathidi, M.: Comparative Analysis of Deep Learning Techniques for Fast Detection of COVID-19 Using CXR Images. In: Advances in Computational Intelligence and Its Applications. pp. 250–258. CRC Press, London (2024). https://doi.org/10.1201/9781003488682-32.

[29] Bani Baker, Q., Hammad, M., Al-Smadi, M., Al-Jarrah, H., Al-Hamouri, R., Al-Zboon, S.A.: Enhanced COVID-19 Detection from X-ray Images with Convolutional Neural Network and Transfer Learning. J Imaging. 10, 250 (2024). https://doi.org/10.3390/jimaging10100250.

[30] Kumar, R.A., Akthar, S.: COVID-19 Chest X-Ray Case Detection with En-semble Deep-Learning. Indian J Sci Technol. 17, 1898–1905 (2024). https://doi.org/10.17485/IJST/v17i18.2220.

[31] Pacal, I.: Improved Vision Transformer with Lion Optimizer for Lung Diseases Detection. Uluslararası Muhendislik Arastirma ve Gelistirme Dergisi. (2024). https://doi.org/10.29137/umagd.1469472.

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